

H. habilis (over 900 cm³), as well as long, slender legs with hip joints well adapted for long-distance walking (Figure 34.50). The fingers were relatively short and straight, suggesting that *H. ergaster* did not climb trees like earlier hominins. *Homo ergaster* fossils have been discovered in far more arid environments than earlier hominins and have been associated with more sophisticated stone tools. Its smaller teeth also suggest that *H. ergaster* either ate different foods than australopiths (more meat and less plant material) or prepared some of its food before chewing, perhaps by cooking or mashing the food.

Homo ergaster marks an important shift in the relative sizes of the sexes. In primates, a size difference between males and females is a major component of sexual dimorphism (see Concept 23.4). On average, male gorillas and orangutans weigh about twice as much as females of their species. In *Australopithecus afarensis*, males were 1.5 times as heavy as females. The extent of sexual dimorphism decreased further in early *Homo*, a trend that continues through our own species: Human males weigh only about 1.2 times as much as females.

The reduced sexual dimorphism may offer some clues to the social systems of extinct hominins. In extant primates, extreme sexual dimorphism is associated with intense male-male competition for multiple females. In species that undergo more pair-bonding (including our own), sexual dimorphism is less dramatic. In *H. ergaster*, therefore, males and females may have engaged in more pair-bonding than earlier hominins did.

Fossils now generally recognized as *H. ergaster* once were considered early members of another species, *Homo erectus*, and some paleoanthropologists still hold this position. *Homo erectus* originated in Africa and was the first hominin



▲ **Figure 34.50 Fossil of *Homo ergaster*.** This 1.7-million-year-old fossil from Kenya belongs to a young *Homo ergaster* male. This individual was tall, slender, and fully bipedal, and he had a relatively large brain.

to migrate out of Africa. The oldest fossils of hominins outside Africa, dating back 1.8 million years, were discovered in the present-day country of Georgia. *Homo erectus* eventually migrated as far as the Indonesian archipelago. Fossil evidence indicates that *H. erectus* became extinct between 200,000 and 70,000 years ago.

Neanderthals

In 1856, miners discovered some mysterious human fossils in a cave in the Neander Valley in Germany. The 40,000-year-old fossils belonged to a thick-boned hominin with a prominent brow. The hominin was named *Homo neanderthalensis* and is commonly called a Neanderthal. Neanderthals were living in Europe by 350,000 years ago and later spread to the Near East, central Asia, and southern Siberia. They had a brain larger than that of present-day humans, buried their dead, and made hunting tools from stone and wood. But despite their adaptations and culture, Neanderthals became extinct at some point between 28,000 and 40,000 years ago.

What is the evolutionary relationship of Neanderthals to *Homo sapiens*? Genetic data indicate that the lineages leading to *H. sapiens* and to Neanderthals diverged about 600,000 years ago (Figure 34.51). This indicates that while Neanderthals and humans share a relatively recent common ancestor, humans did not descend directly from Neanderthals (as was once thought). Neanderthals were, in fact, more closely related to the “Denisovans,” a hominin whose fossils have been discovered in Siberia and Tibet, than to humans.

Another long-standing question is whether mating occurred between Neanderthals and humans, leading to interspecific gene flow. Some researchers have argued that evidence of gene flow can be found in fossils that show

▼ **Figure 34.51 Evolutionary relationships and interbreeding among humans, Neanderthals, and Denisovans.** Genomic data and fossil evidence (see Figure 34.52) document multiple examples of interbreeding among humans, Neanderthals, and Denisovans.

